

Quantifying site-response effect of spatial variability earthquake on seismic failure mode of long-span sea-crossing cable-stayed bridges

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ABSTRACT

Long-span cable-stayed bridge is a prevalent bridge type for sea-crossing passages and occupies an essential role in the transportation system. Non-uniform excitations significantly impact the response of long-span cable-stayed bridges, especially the site-response effect (E_{SR}), which is one of the three primary characteristics. This paper employs a long-span sea-crossing cable-stayed bridge with consideration of hydrodynamic and pile-soil effects to systematically study the influence regularity and parameter sensitivity of the site parameter ω_c on the response of critical components, and develop probabilistic seismic demand models between each component and ω_c . The results reveal that the fragility of each component exhibits dissimilar sensitivities related to ω_c , specifically, the changing amplitude of bearing fragility is greater than that of the bridge pylon. Consequently, when the soil type at the pylons changes from soft to solid, the damage sequence of the main pylon, pier and bearing has been changed, which exhibits different damage mechanism and failure mode. Furthermore, the mathematical model between the seismic parameters and damage mechanism is established to quantify the impact of E_{SR} on the failure mode of sea-crossing cable-stayed bridges. Thus, it could provide a theoretical basis for the earthquake damage assessment and seismic optimization of long-span cable-stayed bridges.

1. Introduction

In the last few decades, many long-span bridges, such as cable-stayed bridges with several advantages, have been widely constructed around China due to the need for the development of regional economics (Ren and Obata, 1999; Chang et al., 2004; Sharabash and Andrawes, 2009; Pang et al., 2014, 2022). Meanwhile, the various types of bridge systems, built or under construction, are inevitably subjected to natural disasters such as earthquakes (Ali and Abdel-Ghaffar, 1994; Ruan-grassamee and Kawashima, 2002; Dyke et al., 2003; Bernknopf and Amos, 2014). The uniform excitation was usually used in the design of seismic resistance and safety assessment of ordinary bridges (Kim et al., 2011; Park and Towashiraporn, 2014), while the same approach was also adopted for such long-span bridges (Valdebenito and Aparicio, 2006), ignoring the influence of the spatial variability of ground motions on the response of bridges (Zanardo et al., 2002; Kuyumcu and Ates, 2012; Zong et al., 2014). This behavior may lead to inconsistencies between the seismic response of structural components and the actual situation, underestimating the adverse effect of spatial variability earthquakes on long-span structures (Hu et al., 2021).

Extensive research has been carried out by scholars on long-span structures considering the spatial variability of earthquakes, including the incoherence effect (E_{IC}), site-response effect (E_{SR}) and wave-passage effect (E_{WP}) (Trifunac and Gicev, 2006; Li et al., 2012, 2013; Zong et al., 2014; Falamarz-Sheikhabadi and Zerva, 2017). Some of these scholars had different views on the effect of E_{IC} for the seismic response of bridges (Quan et al., 2008; Gong et al., 2020; Hu et al., 2021). Quan et al. (2008) pointed out that the amplification of the E_{IC} on the response of structural components should be strictly considered for the safety of bridge structures, and the amplification factors of the displacement and internal force were about 15% and 10%, respectively. Gong et al. (2020) assessed the impact of spatial variability of earthquakes for the safety of the train traversing the multi-tower cable-stayed bridge and found that E_{IC} can impact the train-track-bridge coupled system. Hu et al. (2021) studied the effects of spatial variability parameters for the vulnerability of multi-tower cable-stayed bridge components. The results revealed that the damage probability of the piers and pylons decreases with the increase of the incoherence coefficient ratio. The E_{SR} displayed an amplified influence on the earthquake response of bridge components (Shiravand and Parvanehro, 2019; Gong et al., 2020; Hu et al., 2021).

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Shiravand and Parvanehro (2019) compared the difference in the seismic response of bridge components with uniform and non-uniform excitations. They revealed that the response ratio (the average of the maximum response at non-uniform excitation over that at uniform excitation) decreased as the soil layer changed from solid to soft in all supports. Gong et al. (2020) proposed that the local site effect shall be considered during the analysis of the seismic response of a coupled train-track-bridge system. Hu et al. (2021) concluded that the damage probability of the pylon increased significantly when the soil type at the bottom of the pylon changed from solid soil to soft soil. The previous views were also inconsistent regarding the E_{WP} (Liang et al., 2017; Shiravand and Parvanehro, 2019). Liang et al. (2017) firstly experimentally investigated the effect of E_{WP} on the piles of the cable-stayed bridge. They stated that the earthquake response of bridge components increased simultaneously with the increase of shear wave velocity. Shiravand and Parvanehro (2019) indicated that the effect of wave-passage on the component responses could be negligible.

Considering the spatial variability of earthquakes, a capability-based seismic design approach was suggested to evaluate the seismic behavior of bridges (Kuyumcu and Ates, 2012; Li et al., 2018a, 2018b; Gong et al., 2020; Hu et al., 2021; Wen et al., 2021; Zhong et al., 2022a). Kuyumcu and Ates (2012) suggested that the spatial variability of earthquakes should be considered for calculating the response of long-span bridges. Li et al. (2018a) presented a new approach for generating offshore multi-support ground motions to study and discuss the influence of the soil-structure effects, depth and spatial-varying offshore ground motions, and additional mass of seawater on the bridge system and component vulnerability. Li et al. (2018b) qualitatively evaluated the response of long-span reinforced concrete (RC) bridges at spatial variability earthquakes. They suggested that spatial variability earthquakes should be used as an input for the analysis of the seismic response of bridges on complex terrain. However, few systematic studies and further quantification of the influence of spatial variability parameters on the seismic response and damage sequence of cable-stayed bridge components have been carried out.

Therefore, this paper focuses on the impact of the spatial variability parameters of earthquakes on the damage sensitivity of critical components of long-span sea-crossing cable-stayed bridges. A long-span sea-crossing cable-stayed bridge with consideration of the hydrodynamic and pile-soil effects is employed, and the corresponding three-dimensional (3D) finite element model is built with *OpenSees* software. Based on the IDA method, a group of multi-support ground motion sets is simulated and input with displacement time histories. The seismic vulnerability analysis is conducted to evaluate the damage condition of critical components of cable-stayed bridges, i.e., piers, pylons and bearings, under complex spatial variability ground motions. Finally, due to the differences in sensitivity of critical components to the spatial variability parameters, the effects of spatial variability parameters on the damage mechanism and failure mode of cable-stayed bridges are investigated and quantified.

2. Simulation of spatial variability earthquakes

The method proposed by Konakli and Der Kiureghian (2011, 2012) and adopted by Hu et al. (2021) is applied to generate ground motions considering the spatial variation, which strictly considers three components of spatial effects, i.e., the site-response effect (E_{SR}), wave-passage effect (E_{WP}) and incoherence effect (E_{IC}). Therefore, this section focuses on simulating the displacement and acceleration time histories of the target location based on the acceleration time histories of the existing measured earthquake records and loading them into the finite element model of the cable-stayed bridge in the form of the displacement time history. Note that the simulation method does not consider the influence of the seawater layer on ground motion characteristics considering that it has no significant effect on the horizontal component of the ground motion (Boore and Smith, 1999; Chen et al.,

2015; Li et al., 2015, 2017), and this study is concerned with the horizontal response of the cable-stayed bridge under horizontal loading.

2.1. Theory of generating acceleration and displacement time histories for each position

Step1: Generation of the acceleration time history for the target position m

The self-power spectral density function (SPSD) $G_{mm}(\omega)$ at the target position m can be obtained by the SPSP $G_{cc}(\omega)$ at the reference position c . The SPSP $G_{cc}(\omega)$ of the acceleration process is given by (Zhong et al., 2017, 2018; Hu et al., 2021):

$$G_{cc}(\omega_p) = \frac{N\Delta t}{4\pi} (A_{pc}^2 + B_{pc}^2) = \frac{\Delta t}{\pi N} \left| \sum_{i=1}^N a_c(t_i) \exp(it_i\omega_p) \right|^2 \quad (1)$$

in which $a_c(t_i)$ represents the acceleration time history of the reference location c , N denotes an even number, ω_p denotes the p th order frequency, and Δt denotes the time interval for the time history of the reference location c . The Fourier coefficients A_{pc} and B_{pc} can be calculated according to Konakli and Der Kiureghian (2011, 2012).

Then the SPSP $G_{mm}(\omega)$ at the target position could be obtained by the frequency response function $H(\omega)$ and SPSP $G_{cc}(\omega)$. The corresponding equation can be expressed as:

$$G_{mm}(\omega) = \frac{G_{cc}(\omega) |H_m(\omega)|^2}{|H_c(\omega)|^2} \quad (2)$$

where $H_c(\omega)$ equals

$$H_c(\omega) = \frac{\omega_c^2 + 2i\zeta_c\omega_c\omega}{\omega_c^2 - \omega^2 + 2i\zeta_c\omega_c\omega} \quad (3)$$

in which, ζ_c and ω_c denote the damping ratio and filter frequency at the c th site, respectively. And ω_c can be approximated by first resonant frequency of the soil layer. The reasonable values of these two parameters following Kiureghian and Neuenhofer (1992) and Soylik and Dumanoglu (2004) are adopted and presented in Table 1.

The cross-power spectral density function $G_{cm}(\omega)$ between sites c and m can be expressed as:

$$G_{cm}(\omega) = \gamma_{cm}(\omega) [G_{cc}(\omega) G_{mm}(\omega)]^{\frac{1}{2}} \quad (4)$$

in which $\gamma_{cm}(\omega)$ represents the coherency function between sites c and m . Kiureghian (1996) developed a coherency model for earthquakes after considering the E_{IC} , E_{SR} , and E_{WP} (Luco and Wong, 1986; Konakli and Der Kiureghian, 2011, 2012; Zhong et al., 2017, 2018), which can be expressed as:

$$\begin{aligned} \gamma_{cm}(\omega) &= \gamma_{cm}^{ic}(\omega) \exp[i\theta_{cm}^{wp}(\omega)] \exp[i\theta_{cm}^{sr}(\omega)] \\ &= \exp \left[- \left(\frac{\alpha d_{cm}^w}{v_s} \right)^2 \right] \cdot \exp \left(- i \frac{\omega d_{cm}^w}{v_{app}} \right) \cdot \exp \left\{ - i \tan^{-1} \frac{\text{Im}[H_c(\omega) H_m(-\omega)]}{\text{Re}[H_c(\omega) H_m(-\omega)]} \right\} \end{aligned} \quad (5)$$

in which $\exp[i\theta_{cm}^{sr}(\omega)]$, $\exp[i\theta_{cm}^{wp}(\omega)]$, $\gamma_{cm}^{ic}(\omega)$ represent the site-response, wave-passage and incoherence effects, respectively. And α is an

Table 1
Characteristic parameters of different soil types.

Soil types	ω_c (rad/s)	ζ_c	v_s (m/s)
Soft	2	0.1	100
Medium soft	5	0.2	150
Medium	10	0.4	250
Medium solid	15	0.6	500
Solid	20	0.8	800

incoherence coefficient, d_{cm} and d_{cm}^L denote the distance between the positions c and m and the projected distance of the wave propagating from position c to m , respectively, v_s and v_{app} denote the velocity of the shear wave for corresponding soil medium and the apparent wave velocity, respectively. Note that v_{app} yields v_s if the example bridge is far away from the epicenter (Kiureghian, 1996; Zhong et al., 2017, 2018). Referring to earlier works (Bi et al., 2011; Hu et al., 2021), the value of v_s corresponding soil types is presented in Table 1. In addition, the reasonable values of the ratio of the incoherence coefficient β ($\beta = \alpha/v_s$) were $2.0 \times 10^{-4} \text{ s/m} \sim 10.0 \times 10^{-4} \text{ s/m}$ following the previous research (Luco and Wong, 1986). The Fourier series can represent the acceleration time history of any site c because it is a random process (Konakli and Der Kiureghian, 2011, 2012).

$$a_c(t_i) = A_{0c} + \sum_{p=1}^{N-1} [A_{pc} \cos(\omega_p t_i) + B_{pc} \sin(\omega_p t_i)] + (-1)^i A_{(N/2)c} \quad c = 1, 2, \dots, N \quad (6)$$

in which N denotes the number of discrete observations that are sampled with equal time intervals.

Consider M discrete spatial positions of a Gaussian, zero-mean, and homogenous ground motion random site. The zero-mean Fourier coefficient set with the frequency ω_p is $X_p = [X_{p1}, X_{p2}]^T$, in which X_{p1} and X_{p2} denote Fourier coefficients of the reference and target positions, respectively. X_{p1} and X_{p2} can be expressed as:

$$\begin{aligned} X_{p1} &= (A_{1c}, B_{1c}, A_{2c}, B_{2c}, \dots, A_{nc}, B_{nc})^T \\ X_{p2} &= (A_{(n+1)c}, B_{(n+1)c}, A_{(n+2)c}, B_{(n+2)c}, \dots, A_{Mc}, B_{Mc})^T \end{aligned} \quad (7)$$

The covariance matrix of these coefficients can be expressed as:

$$\Sigma_{pp} = \begin{bmatrix} \Sigma_{pp,11} & \Sigma_{pp,12} \\ \Sigma_{pp,21} & \Sigma_{pp,22} \end{bmatrix}_{2n \times 2n} \quad (8)$$

in which $\Sigma_{pp,12(21)}$ denotes the covariance between $X_{p1(2)}$ and $X_{p2(1)}$. The corresponding covariance matrix and mean equal

$$\begin{aligned} M_{(p,2|1)} &= \Sigma_{pp,21} (\Sigma_{pp,11})^{-1} X_{p1} \\ \Sigma_{(p,2|1)} &= \Sigma_{pp,22} - \Sigma_{pp,21} (\Sigma_{pp,11})^{-1} \Sigma_{pp,12} \end{aligned} \quad (9)$$

Then the Fourier coefficient set of the target position m can be expressed as:

$$X_{p2} = M_{(p,2|1)} + L_{(p,2|1)}^T z_p \quad (10)$$

in which, z_p denotes a vector of $2(n-1)$ uncorrelated standard normal variate, $L_{(p,2|1)}^T$ denotes the up-triangle matrix. The time histories of the stationary acceleration at target location m can be obtained by applying X_{p2} to Eq. (6).

Finally, the time histories of the non-stationary acceleration at target location m could be determined from multiplying the time history of the stationary acceleration with an envelope function, which is expressed as (Bi and Hao, 2012):

$$\zeta(t) = \begin{cases} (t/t_0)^2 & 0 \leq t \leq t_0 \\ 1 & t_0 < t \leq t_n \\ \exp[-0.155(t - t_n)] & t_n < t \leq T \end{cases} \quad (11)$$

in which, the start time of the stable period of earthquake $t_0 = 2 \text{ s}$, the end time $t_n = 10 \text{ s}$, and T denotes its total duration (Bi and Hao, 2012).

Step2: Transformation of acceleration into displacement time history

The displacement time history is generally acquired as input motion for the multi-support excitation analysis of the bridge model in *OpenSees*. Therefore, a function proposed by Boore et al. (2002) is adopted to obtain the displacement time histories at reference and target positions, which guarantees that both the residual displacement and the initial

acceleration are zero. The approach of transforming acceleration into displacement time history can be found in Boore et al. (2002) and Liao and Zerva (2006).

2.2. Selection of input ground motions and cases simulation

In view of the limited ground motion records at the geographical location of the bridge, fifty pairs of earthquake records were randomly selected for IDA in the Pacific Earthquake Engineering Research Center (PEER) (Zhong et al., 2022c) with the following criteria: a) the peak ground velocity (PGV) is between 0.11 and 1.48 m/s; b) the magnitude (M) is between 5.8 and 7.62; and c) the source distance (R) is between 0.07 and 28.04 km. These records cover a wide range of site types and fault mechanisms and serve as ground motion inputs for the reference position S1 (the left side pier of the example bridge), whose detailed information is given in Appendix Table A. The detailed locations can be illustrated in Fig. 1. Note that the positions S1, A1, P1, P2, A2 and S2 refer to the base of side pier No. 1, auxiliary pier No. 1, pylon No. 1, pylon No. 2, auxiliary pier No. 2, and side pier No. 2 of the example bridge, respectively. The acceleration time histories of the target positions A1, P1, P2, A2 and S2 can be calculated from the acceleration time histories of the reference position S1. These acceleration time histories are then transformed into the displacement time histories (Boore et al., 2002; Liao and Zerva, 2006). Finally, the displacement time histories at six different supports are input to the bridge model along the longitudinal direction of the bridge.

Taking into account the actual location of the example bridge, the foundations of the piers located on the beach are assumed to be solid, while the foundations of the pylons are flexible. Therefore, this paper considers the variation of soil conditions at P1 and P2 to study the influence of the spatial variability parameters of earthquakes on critical components of cable-stayed bridges, and establishes a set of cases (Case 1–5: $\omega_c = 20, 15, 10, 5, 2 \text{ rad/s}$) under the condition of the horizontal peak ground velocity $V_x = 0.1\text{--}1.2 \text{ m/s}$, as shown in Table 2. Figs. 2 and 3 illustrate the acceleration and displacement time histories at P1 under the condition of $V_x = 0.1 \text{ m/s}$ and $\omega_c = 2\text{--}20 \text{ rad/s}$, respectively. Fig. 4 illustrates the displacement time histories of different positions under the condition of $V_x = 0.1 \text{ m/s}$ and $\omega_c = 20 \text{ rad/s}$.

3. Case study of the cable-stayed bridge

3.1. Bridge modelling

This study adopts a long-span sea-crossing cable-stayed bridge with four reinforced concrete piers and two reinforced concrete pylons as a numerical example, located on the southeast coast of China. The overall layout of this example bridge is illustrated in Fig. 5(a). It consists of a 532 m main span and two 328 m symmetric side spans arranged with auxiliary piers. The steel truss girder is connected to pylons with 136 cables arranged in a fan-shaped configuration. The side pier and auxiliary pier on the beach are attached to foundations and used as fixed supports with a height of 50 m. The height of the pylon is 220 m. It is supported by a pile group foundation, which consists of 18 piles with a diameter of 4.5 m and a length of 47 m. Fig. 5(b) illustrates the cross sections of piers and pylons and their reinforcement percentages.

A 3D finite element bridge model simulated using the *OpenSees* software is shown in Fig. 5. Generally, the steel truss girder is not considered for its occurrence of damage under earthquakes. Thus, it is simulated with the elastic beam-column element. The cables connecting the main girder to the pylon are modeled with truss elements, and Young's modulus is modified by the Ernst formula (Hu et al., 2021). The pylons, piers and piles are modeled using 3D nonlinear beam-column elements with four integration points and fiber sections. The corresponding cross sections of the pile, pylon and pier are modeled using longitudinal reinforcement, unconfined concrete and confined concrete fibers. As depicted in Fig. 5(c), the former is defined with the Steel01

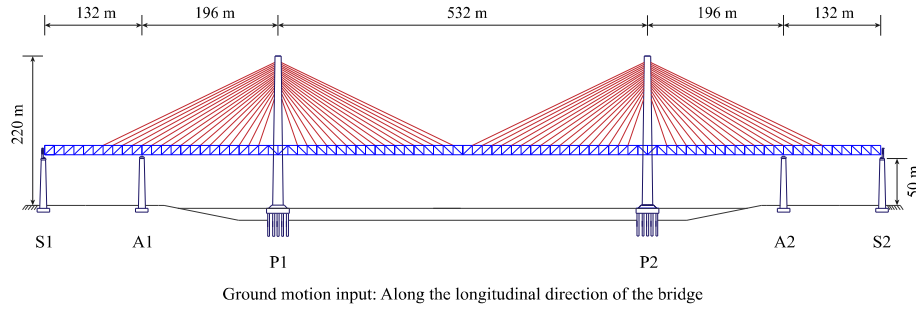


Fig. 1. Illustration of the location and direction of ground motion inputs.

Table 2

Cases simulation considering the site-response effect (E_{SR}).

Cases ($C_{i,j}$)	Soil types (ω_c : rad/s)				
	$\omega_c = 20$	$\omega_c = 15$	$\omega_c = 10$	$\omega_c = 5$	$\omega_c = 2$
$V_x = 0.1$ m/s	$C_{1,1}$	$C_{2,1}$	$C_{3,1}$	$C_{4,1}$	$C_{5,1}$
$V_x = 0.4$ m/s	$C_{1,2}$	$C_{2,2}$	$C_{3,2}$	$C_{4,2}$	$C_{5,2}$
$V_x = 0.8$ m/s	$C_{1,3}$	$C_{2,3}$	$C_{3,3}$	$C_{4,3}$	$C_{5,3}$
$V_x = 1.2$ m/s	$C_{1,4}$	$C_{2,4}$	$C_{3,4}$	$C_{4,4}$	$C_{5,4}$

model and the latter is defined with Concrete01 model in *OpenSees* (Yang et al., 2023). The yield stress and Young's modulus of the longitudinal reinforcements are 400 MPa and 200 GPa. The peak compressive strengths and peak strains of the unconfined and confined concrete are 38.5 MPa, 43.4 MPa and 0.002, 0.0022, respectively. The link elements with a bilinear material are used to model the friction pendulum bearings (Zhong et al., 2023).

In addition, the piles of the piers can remain elastic under earthquakes considering they are located on the beach, so the simplified 6×6 soil springs are used to simulate the pile-soil effect of the pile

foundations of the piers (Li et al., 2018a). The nonlinear p-y spring is employed to model the pile-soil interaction for the pile group foundation of pylons, where the values of the p-y curve can be obtained based on the equation presented in API (API, 1993). In addition, the fluid-structure interaction is simulated for the sub-structures (piles) in seawater by adding hydrodynamic additional mass at the nodes (Wei et al., 2021), as shown in Fig. 6.

3.2. The validation of the numerical model

The moment-curvature curves of sections at the bottom of the pier and pylon are obtained by the pushover analysis of the cable-stayed bridge model along the longitudinal direction, as illustrated in Fig. 7, and the results are compared with the moment-curvature curves at the corresponding locations calculated by Xtract software (Chadwell and Imbsen, 2004), respectively. In addition, Table 3 shows the comparison of the first five order natural vibration periods of the cable-stayed bridge model established by both OpenSees and ANSYS software. The results are relatively consistent.

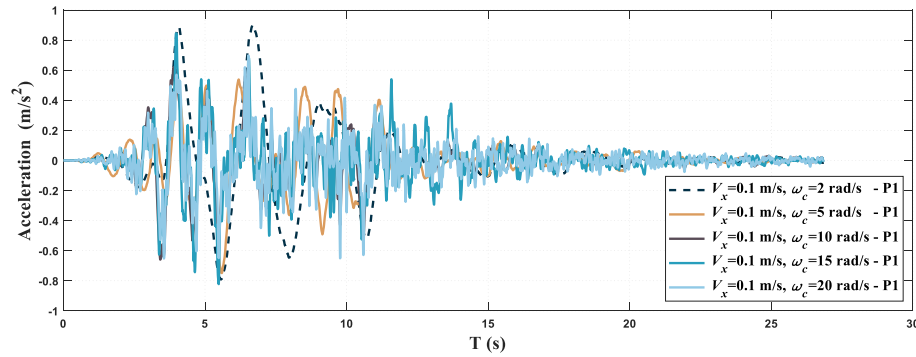


Fig. 2. Acceleration time histories at P1 considering different soil types.

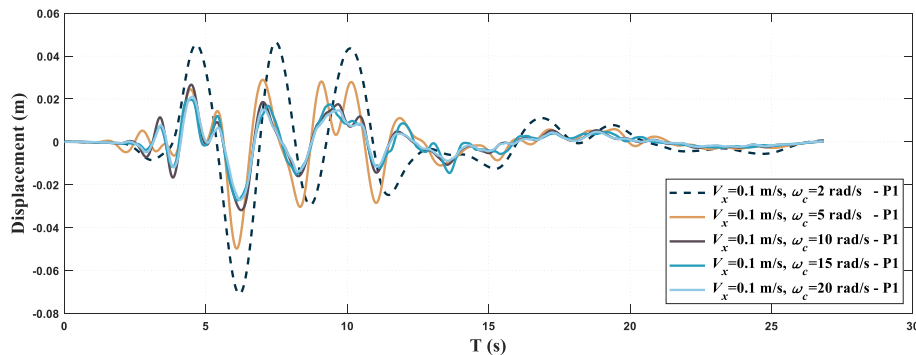


Fig. 3. Displacement time histories at P1 considering different soil types.

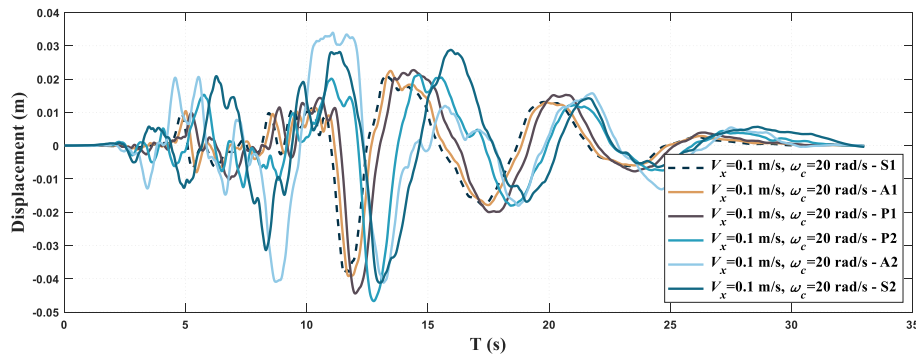


Fig. 4. Displacement time histories at each location under the condition of $V_x = 0.1$ m/s.

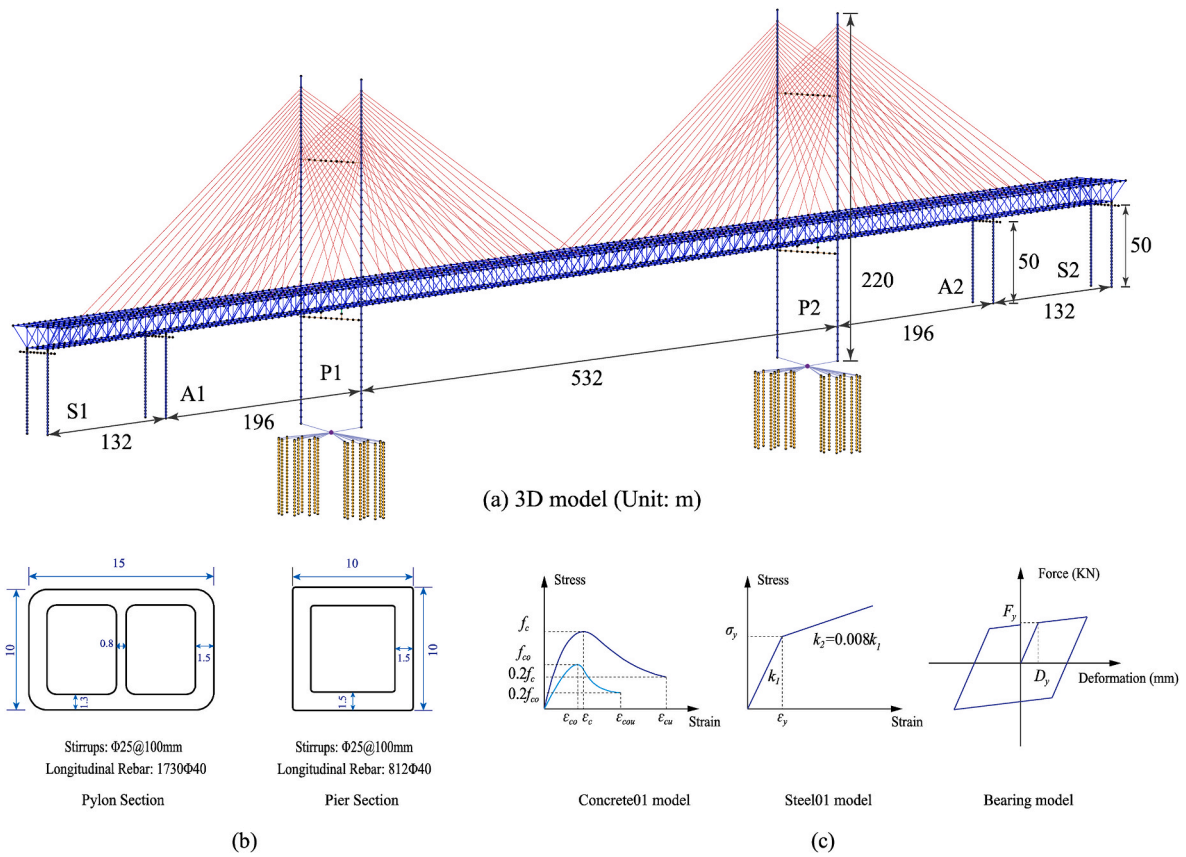


Fig. 5. Overview of the sea-crossing cable-stayed bridge.

3.3. Intensity Measures and capacity definition

The selection of the optimal Intensity Measure (IM) can precisely predict the structural seismic response to assess its damage probability accurately. For cable-stayed bridges, the reasonable selection of IM has been studied extensively by previous researchers (Guo et al., 2020; Li et al., 2017, 2018a; Ma et al., 2019; Wei et al., 2020; Zhong et al., 2019). Among them, Wei et al. (2020) and Zhong et al. (2019) stated that the peak ground velocity (PGV) was a selection of the optimal IM for long-span cable-stayed bridges with regard to efficiency, practicality, adequacy, proficiency, etc. Moreover, Yang et al. (2009) also showed that the PGV was suitable for medium-period and long-period structures. Therefore, PGV is used as IM to observe the seismic response and damage condition of cable-stayed bridge components.

The curvature ductility of auxiliary piers No. 1 and 2, side piers No. 1 and 2, and pylons No. 1 and 2, and the displacements of friction

pendulum bearings at each location are adopted as the engineering demand parameters of cable-stayed bridges in this paper (Zhong et al., 2022b; Pang et al., 2022; Chen et al., 2022). Meanwhile, according to the scholars' research (HAZUS99-SR2., 1999; Pan et al., 2010a, b), the damage states of structural components could be classified into four levels: Slight Damage (SD), Moderate Damage (MD), Extensive Damage (ED) and Collapse Damage (CD).

The limit states of friction pendulum bearings are defined following the work of Jiao (2008). The definition of the capacity limit states of the piers and pylons of the example bridge can follow Ramanathan (2012). The corresponding values are shown in Table 4.

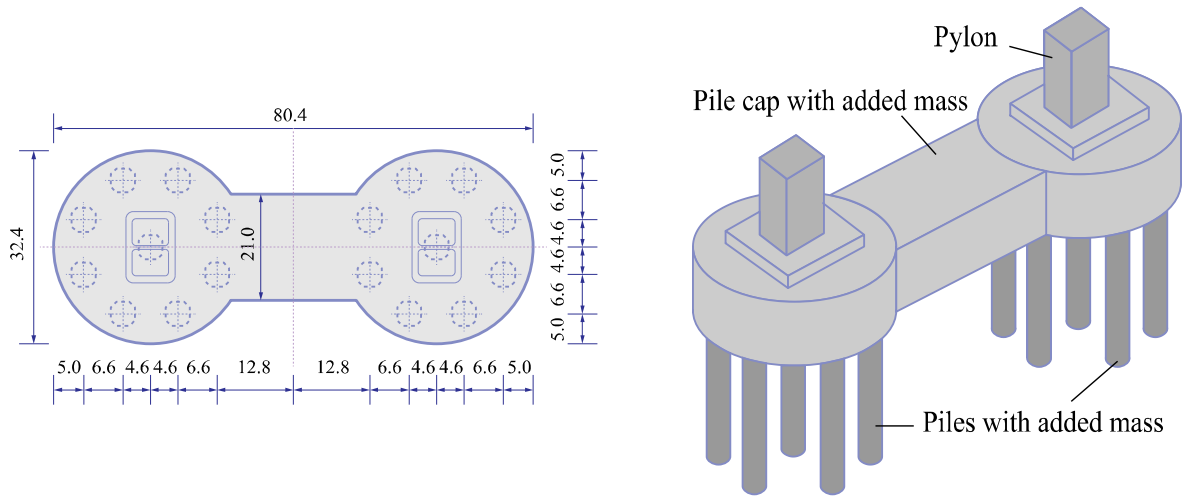


Fig. 6. Pile group foundation considering hydrodynamic effects.

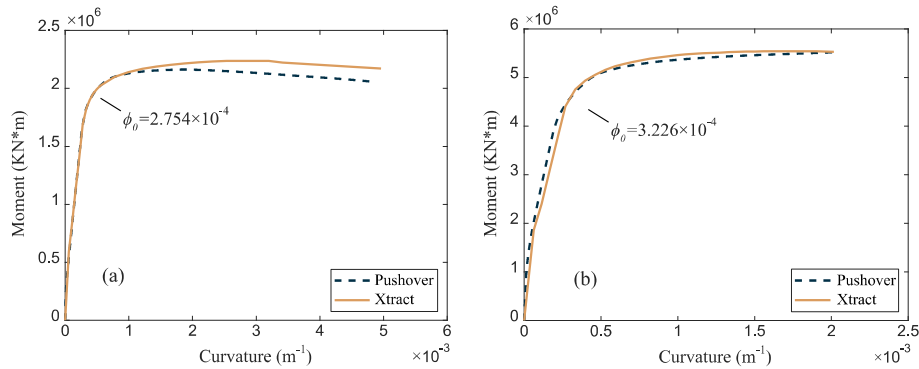


Fig. 7. Comparison of the moment-curvature curves under the pushover analysis and calculation by Xtract for (a) the bottom of the pier, (b) the bottom of the pylon.

Table 3

The first five order natural vibration periods.

Order	Modeling with the OpenSees software	Modeling with the ANSYS software
1	5.54 s	5.44 s
2	3.40 s	3.38 s
3	3.09 s	3.25 s
4	3.06 s	3.17 s
5	2.35 s	2.29 s

Table 4

Median of capacity for different components.

Components	SD	MD	ED	CD
Curvature ductility of pylons (φ_{c1})	1.00	2.00	4.00	7.00
Curvature ductility of piers (φ_{c2})	1.00	2.00	4.00	7.00
The displacement of bearings at the pylon [δ_{d1} (m)]	0.50	1.00	1.30	1.60
The displacement of bearings at the pier [δ_{d2} (m)]	0.15	0.30	0.50	0.80

4. Methodology of fragility analysis

4.1. Fragility function

The fragility analysis method is commonly employed to assess the response performance of bridge components under earthquake loading (Zhong et al., 2020). The component fragility curve is a representation of the conditional probability that the seismic response of a component exceeds the limit state at a certain earthquake intensity (IM) (Pang et al.,

2014). The corresponding function is as follows:

$$P_f = P[\mu_d \geq \mu_c | IM] = \Phi \left(\frac{\ln(\mu_d) - \ln(\mu_c)}{\sqrt{\sigma_d^2 + \sigma_c^2}} \right) \quad (12)$$

in which σ_d and μ_d denote the lognormal standard deviation and median of seismic demand, respectively, σ_c and μ_c denote the lognormal standard deviation and median of capacity, respectively.

4.2. Demand models considering site-response effect of spatial variability earthquakes

The probabilistic seismic demand model describes a relationship between the ground motion intensity (IM) and the structural seismic demand, and the seismic demand model μ_d is usually defined as a linear regression model for $\ln(IM)$ - $\ln(\mu_d)$ pairs obtained from the simulations (Cornell et al., 2002; Yang et al., 2023)

$$\mu_d = aIM^b \quad (13)$$

in which, a and b are regression coefficients. The σ_d can be calculated by Eq. (14):

$$\sigma_d = \sqrt{\frac{\sum [\ln(\mu_i) - \ln(\mu_d)]^2}{n - 2}} \quad (14)$$

where, n is the number of seismic response analysis, μ_i denotes the demand for the i th seismic response analysis.

The seismic responses of each component were recorded in the time

history analysis: the displacements of bearings at pylons No. 1–2, side piers No. 1–2 and auxiliary piers No. 1–2 (δ_{d-P1} , δ_{d-P2} , δ_{d-S1} , δ_{d-S2} , δ_{d-A1} and δ_{d-A2} , in meters) and the curvature ductility at positions S1, A1, P1, P2, A2, and S2 (ϕ_{d-S1} , ϕ_{d-A1} , ϕ_{d-P1} , ϕ_{d-P2} , ϕ_{d-A2} and ϕ_{d-S2}). Referring to Eq. (13), the demand models of piers, pylons and bearings can be determined based on the component responses and V_x , and the two-dimensional (2D) demand models of P1 at each case ($\omega_c = 2, 5, 10, 15, 20$ rad/s) are presented in Fig. 8(a). Table 5 presents the regression coefficients of 2D demand models of some components in each case. In addition, the relationship between component responses and ω_c under the condition of $V_x = 1.2$ m/s is explored, and the fitted curves are shown in Fig. 8(b). It can be observed that the model fits well with the data points and meets the requirement of the study. Based on these results, the expression considering V_x and ω_c can be given as $\mu_d = \gamma_1 V_x^{\gamma_2} (R + 1)$, where the amplification factor R is set to take the site-response effect (ω_c) into account in Eq. (13). Thus, the 3D demand models can be expressed as:

$$\mu_d = \gamma_1 V_x^{\gamma_2} ((\gamma_3 e^{-\gamma_4 \omega_c} + \gamma_5) + 1) \quad (15)$$

in which, γ_i ($i = 1, 2, 3, 4, 5$) are the fitted regression coefficients. Eq. (15) is satisfied with μ_d determined by V_x regardless of the influence of the site-response effect. Fig. 8(c) and Table 6 present the 3D demand models and regression coefficients for 3D demand models of critical components, respectively. The standard dispersion for the regression surface can be given as (Pan et al., 2010b):

$$\sigma_d = \sqrt{\frac{\sum [\ln(\mu_d) - \ln(\mu_i)]^2}{n - 3}} \quad (16)$$

Table 5

Regression coefficients of 2D demand models for some components of each case.

Components	Cases	a	b	σ_d	R^2
Pylon No. 1	$C_{1, 1-4}$	1.02	1.05	1.37	0.85
	$C_{2, 1-4}$	1.17	1.07	1.35	0.83
	$C_{3, 1-4}$	1.29	1.06	1.35	0.84
	$C_{4, 1-4}$	1.56	1.06	1.34	0.80
	$C_{5, 1-4}$	1.95	1.08	1.33	0.80
Assistant Pier No. 1	$C_{1, 1-4}$	1.47	0.72	0.67	0.82
	$C_{2, 1-4}$	1.54	0.71	0.65	0.80
	$C_{3, 1-4}$	1.59	0.72	0.66	0.80
	$C_{4, 1-4}$	1.71	0.75	0.64	0.78
	$C_{5, 1-4}$	1.77	0.74	0.64	0.77
Side Pier No. 1	$C_{1, 1-4}$	1.49	0.70	0.67	0.84
	$C_{2, 1-4}$	1.59	0.74	0.65	0.81
	$C_{3, 1-4}$	1.69	0.73	0.65	0.82
	$C_{4, 1-4}$	1.86	0.77	0.64	0.79
	$C_{5, 1-4}$	2.04	0.77	0.61	0.76
Bearing at Pylon No. 1	$C_{1, 1-4}$	0.45	0.80	0.38	0.96
	$C_{2, 1-4}$	0.49	0.82	0.37	0.94
	$C_{3, 1-4}$	0.54	0.84	0.35	0.89
	$C_{4, 1-4}$	0.75	0.86	0.33	0.85
	$C_{5, 1-4}$	1.21	0.85	0.34	0.87
Bearing at Assistant Pier No. 1	$C_{1, 1-4}$	0.54	0.80	0.46	0.91
	$C_{2, 1-4}$	0.62	0.83	0.44	0.93
	$C_{3, 1-4}$	0.67	0.83	0.44	0.90
	$C_{4, 1-4}$	0.81	0.85	0.41	0.84
	$C_{5, 1-4}$	1.21	0.86	0.43	0.83
Bearing at Side Pier No. 1	$C_{1, 1-4}$	0.48	0.79	0.37	0.90
	$C_{2, 1-4}$	0.56	0.82	0.38	0.95
	$C_{3, 1-4}$	0.61	0.84	0.34	0.92
	$C_{4, 1-4}$	0.69	0.85	0.31	0.88
	$C_{5, 1-4}$	0.96	0.87	0.29	0.86

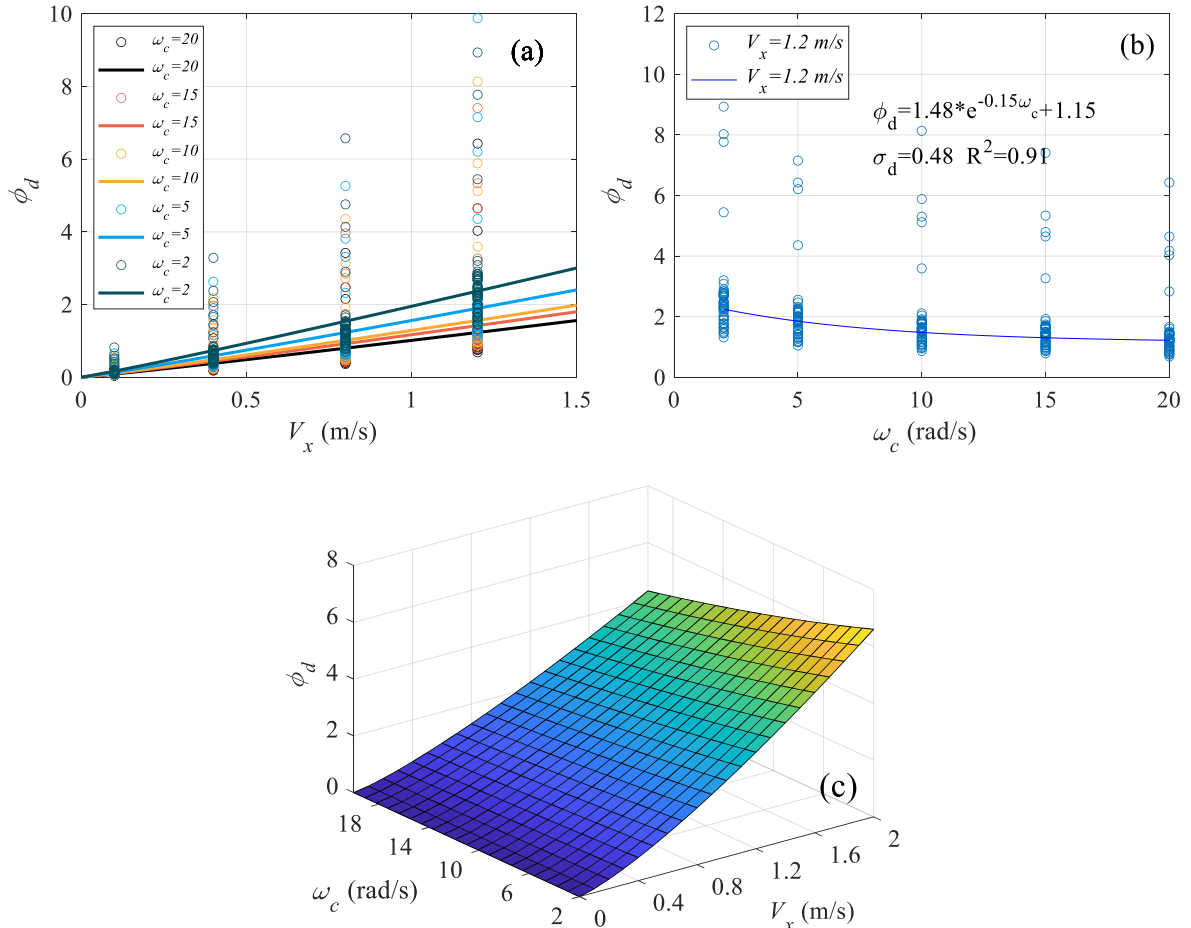
**Fig. 8.** Demand models of P1 considering V_x and ω_c .

Table 6

Regression coefficients of 3D demand models for each component.

Components	γ_1	γ_2	γ_3	γ_4	γ_5
Pylon No. 1 (φ_{d-P1})	0.95	1.30	2.87	0.03	-0.90
Pylon No. 2 (φ_{d-P2})	0.91	1.30	2.81	0.03	-0.82
Assistant Pier No. 1 (φ_{d-A1})	0.76	0.91	1.05	0.02	0.52
Assistant Pier No. 2 (φ_{d-A2})	1.78	0.91	2.73	0.004	-2.67
Side Pier No. 1 (φ_{d-S1})	1.19	0.91	2.81	0.01	-2.05
Side Pier No. 2 (φ_{d-S2})	9.13	0.91	2.52	8.17×10^{-4}	-3.31
Bearing at Pylon No. 1 (δ_{d-P1})	0.51	0.97	0.87	0.13	-0.14
Bearing at Pylon No. 2 (δ_{d-P2})	0.46	1.04	1.77	0.04	-1.01
Bearing at Assistant Pier No. 1 (δ_{d-A1})	0.58	0.89	1.26	0.03	0.70
Bearing at Assistant Pier No. 2 (δ_{d-A2})	1.30	0.84	1.60	0.01	-1.95
Bearing at Side Pier No. 1 (δ_{d-S1})	4.37	1.01	1.42	0.002	-2.25
Bearing at Side Pier No. 2 (δ_{d-S2})	4.27	1.21	1.43	0.002	-2.27

5. Site-response effect on failure mode of cable-stayed bridges

5.1. Effect of E_{SR} on fragility curves of components

Choosing PGV as IM and using the demand model and capacity model of each component, the fragility curves of pylons, piers and bearings could be easily determined by Eq. (12) under the action of complex spatial variability ground motions. Figs. 9–14 depict the fragility curves of each component for different cases under slight and moderate damage states. The median PGV is adopted to compare and analyze the relative vulnerability between bridge components under the same conditions (Zhong et al., 2022b). In addition, note that BP1-2, BS1-2 and BA1-2 denote the bearings at pylons No. 1–2, side piers No. 1–2 and auxiliary piers No. 1–2, respectively.

It can be observed from Figs. 9–14 that the probability of damage to bearings, pylons and piers gradually increases as the value of ω_c decreases. Among them, the magnitude of change of the damage probability for auxiliary piers No. 1–2 and side piers No. 1–2 is smaller than that of pylons No. 1–2. The reason for this phenomenon is that the change from solid to soft soil occurs at the pylons, so the probability of damage to pylons increases rapidly. The side piers and auxiliary piers are also affected to varying degrees.

For side piers No. 1–2 (Figs. 11 and 12), the component containing a 50% probability of exceeding *SD* and *MD* is always located first at the

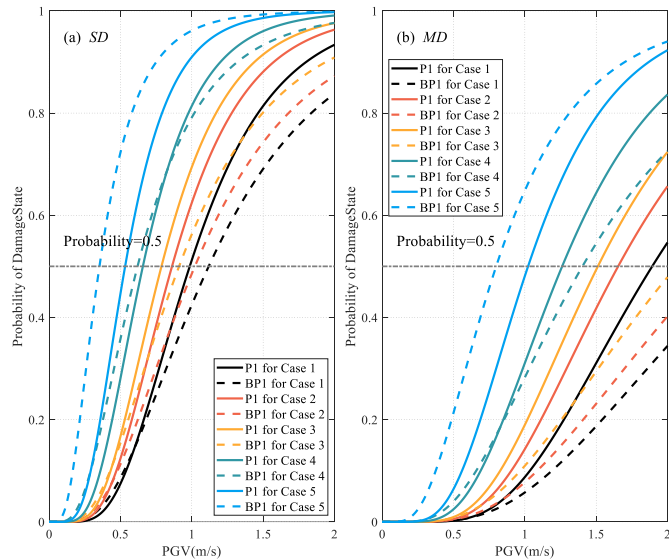


Fig. 9. Comparison of the damage-exceedance probability of P1 and BP1 for each case under (a) *SD*, (b) *MD*.

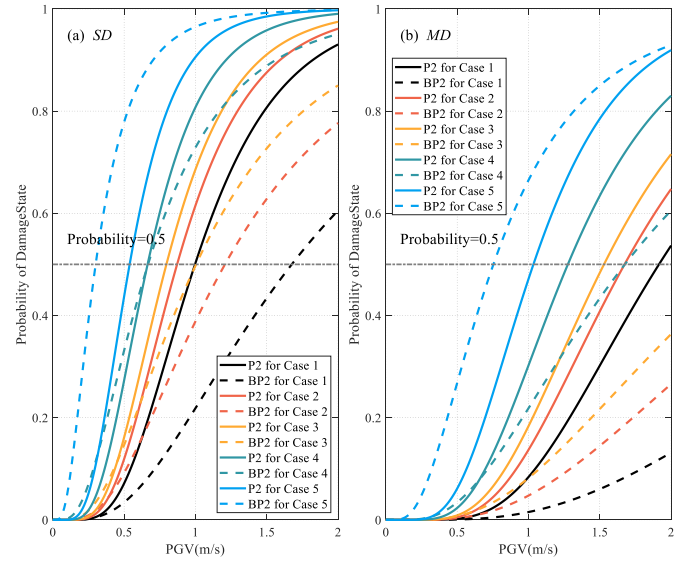


Fig. 10. Comparison of the damage-exceedance probability of P2 and BP2 for each case under (a) *SD*, (b) *MD*.

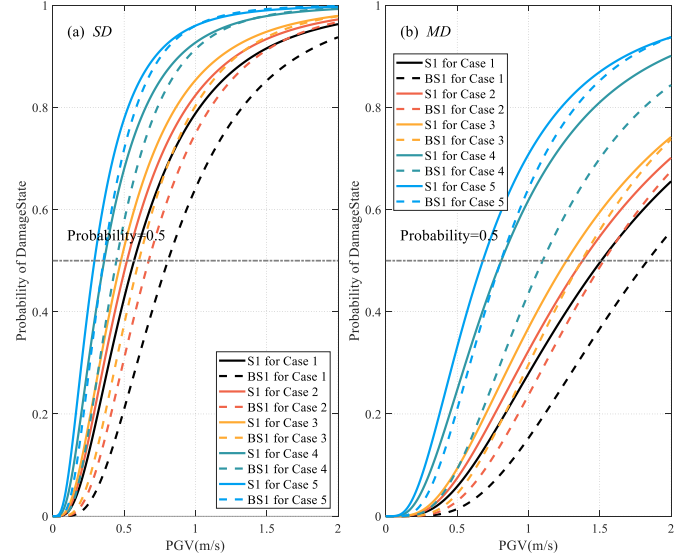


Fig. 11. Comparison of the damage-exceedance probability of S1 and BS1 for each case under (a) *SD*, (b) *MD*.

bottom of piers in Cases 1–5, that is, the side piers are more prone to damage than bearings at side piers. While for pylons No. 1–2 (Figs. 9 and 10), the pylons' bearings would be damaged at first before the bottom of pylons under *SD* and *MD* in Case 5 ($\omega_c = 2$ rad/s). With the decrease of ω_c , the situation turns. The bottom of pylons would be damaged at first before bearings at pylons under *SD* and *MD* in Case 1 ($\omega_c = 20$ rad/s). Similar results for auxiliary piers No. 1–2 can also be observed from Figs. 13 and 14, the bearings at auxiliary piers would be damaged at first before auxiliary piers under *SD* and *MD* in Case 5, whereas in Case 1, the auxiliary piers are more prone to damage than their bearings. This phenomenon indicates that differences in the sensitivity of different components of the example bridge to the site parameter ω_c under the action of ground motions considering the spatial variability. After the soil type of the pylon becomes solid from soft, the original damage mechanism changes accordingly, causing auxiliary piers and pylons to become vulnerable components.

After the above discussion of the damage sequence of auxiliary piers,

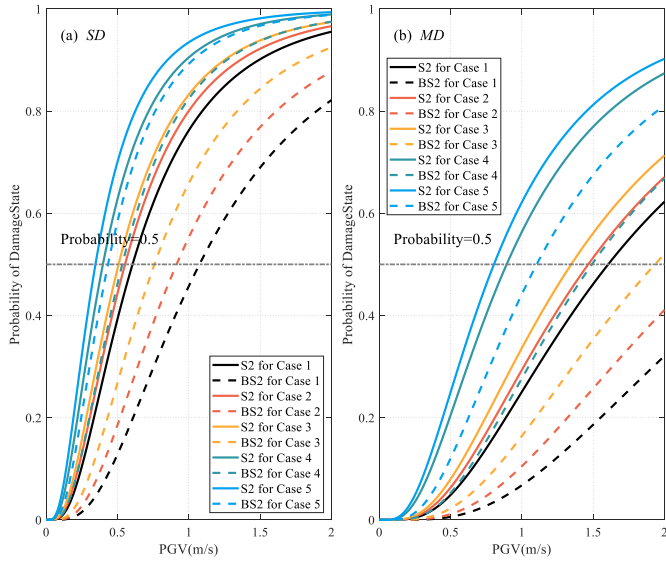


Fig. 12. Comparison of the damage-exceedance probability of S2 and BS2 for each case under (a) SD, (b) MD.

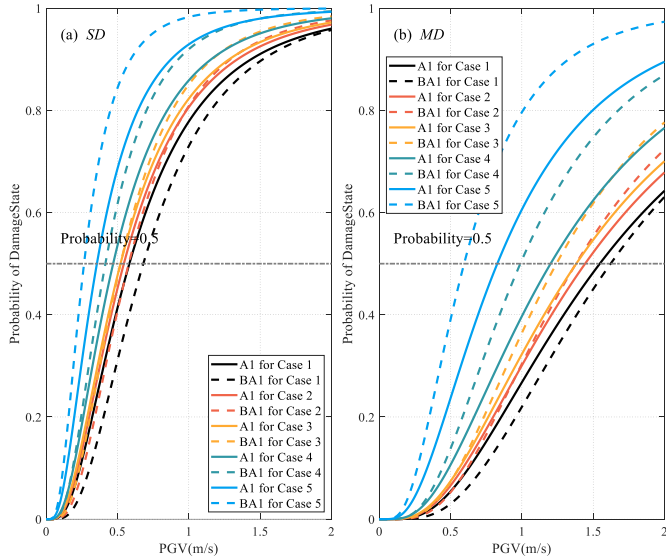


Fig. 13. Comparison of the damage-exceedance probability of A1 and BA1 for each case under (a) SD, (b) MD.

pylons and bearings under longitudinal earthquakes, the damage direction and location of the cable-stayed bridge with each case are summarized and depicted in Figs. 15 and 16. The components in the red area indicate that they are in the longitudinal damage state, i.e., the vulnerable components in that case.

5.2. Effect of E_{SR} on fragility surfaces of components

The function for the seismic fragility surface of critical components, considering the spatial variability of ground motions can be expressed as (Pan et al., 2010b; Zhong et al., 2022b):

$$P_f = P[\mu_d \geq \mu_c | V_x, \omega_c] = \Phi \left(\frac{\ln(\mu_d) - \ln(\mu_c)}{\sqrt{\sigma_d^2 + \sigma_c^2}} \right) \quad (17)$$

$$= \Phi \left(\frac{\ln[\gamma_1 V_x^{\gamma_2} (\gamma_3 e^{-\gamma_4 \omega_c} + \gamma_5) + 1] - \ln(\mu_c)}{\sqrt{\sigma_d^2 + \sigma_c^2}} \right)$$

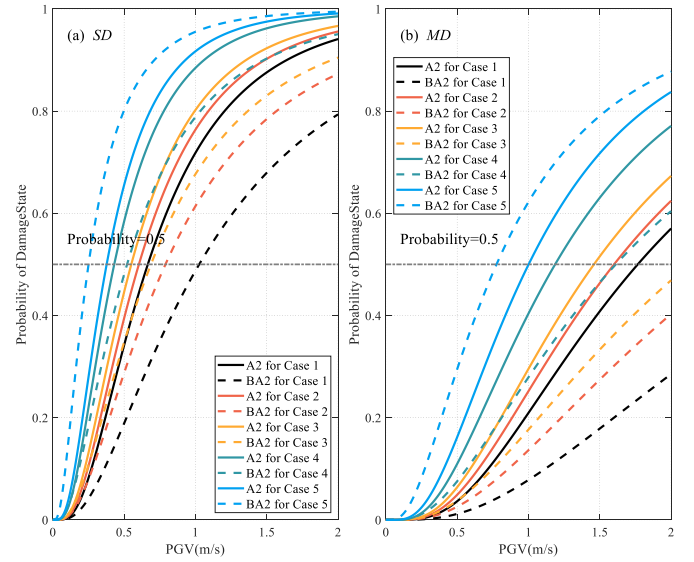


Fig. 14. Comparison of the damage-exceedance probability of A2 and BA2 for each case under (a) SD, (b) MD.

To better observe the influence of spatial variability parameters on the damage condition of critical components, the seismic fragility surfaces of each component corresponding to SD and MD are developed by Eq. (17). Figs. 17(a) and 18(a) show the damage probabilities of pylon No. 1 and the bearing at pylon No. 1 decrease from 0.96 to 0.91 and 0.98 to 0.72 in SD, respectively, as ω_c increases with V_x equals 2.0 m/s. Figs. 17(b) and 18(b) show the damage probabilities of pylon No. 1 and the bearing at pylon No. 1 decrease from 0.86 to 0.76 and 0.88 to 0.48 in MD, respectively, as ω_c increases with V_x equals 2.0 m/s. This indicates that when V_x reaches a specific value, the damage probability of the pylon No. 1 gradually exceeds the damage probability of the bearing at pylon No. 1 as ω_c increases.

In addition, as observed in Figs. 19 and 20, at V_x equal to 2.0 m/s, the damage probabilities of pylon No. 2 and the bearing at pylon No. 2 in SD decrease from 0.97 to 0.91 and 0.99 to 0.66, respectively, with the increase of ω_c ; the damage probabilities of pylon No. 2 and the bearing at pylon No. 2 in MD decrease from 0.86 to 0.75 and 0.92 to 0.41, respectively, with the increase of ω_c . Similarly, this phenomenon shows that the damage probability of the pylon No. 2 gradually exceeds that of the bearing at pylon No. 2 as ω_c increases.

5.3. Effect of E_{SR} on damage mode

As observed in the fragility analysis of the cable-stayed bridge in Sections 5.1 and 5.2, variations in soil types at the pylons adversely affect the critical components of the bridge, meanwhile the damage mechanism of the cable-stayed bridge is changed given that the damage to each component exhibits dissimilar sensitivities associated with site parameter ω_c and ultimately leads to the failure of the bridge. And this section focuses on quantifying the impact of ω_c for the failure mode of the example bridge.

The difference value of the damage probability between the pylon No. 1 and the bearing at this location under the joint participation of V_x and ω_c can be obtained with the following equation

$$P_{f-DP1}(V_x, \omega_c) = P_{f-P1}(V_x, \omega_c) - P_{f-BP1}(V_x, \omega_c) \quad (18)$$

in which, P_{f-P1} and P_{f-BP1} can be calculated by Eq. (17). On the basis of the difference value P_{f-DP1} , two regions divided by $P_{f-DP1} = 0$ associated with SD and MD are developed as depicted in Fig. 21(a) and (b), respectively. Note that region R_{BP1} indicates that $P_{f-DP1} < 0$, i.e., the bearing of pylon No. 1 is more vulnerable than pylon No. 1 under the

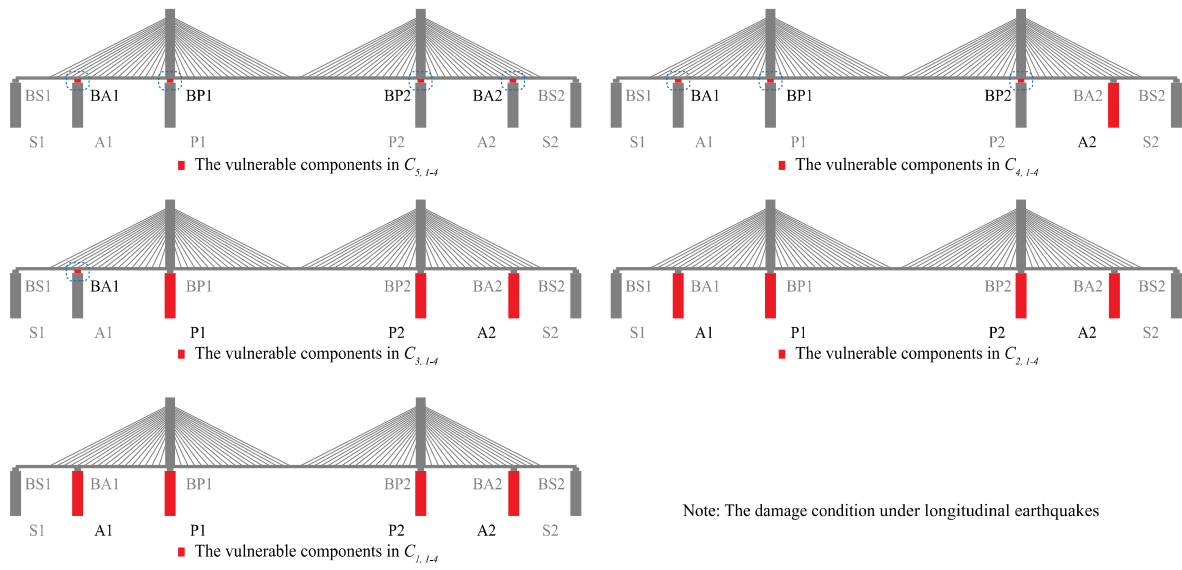


Fig. 15. Illustration of the change in location of vulnerable components for each case under SD.

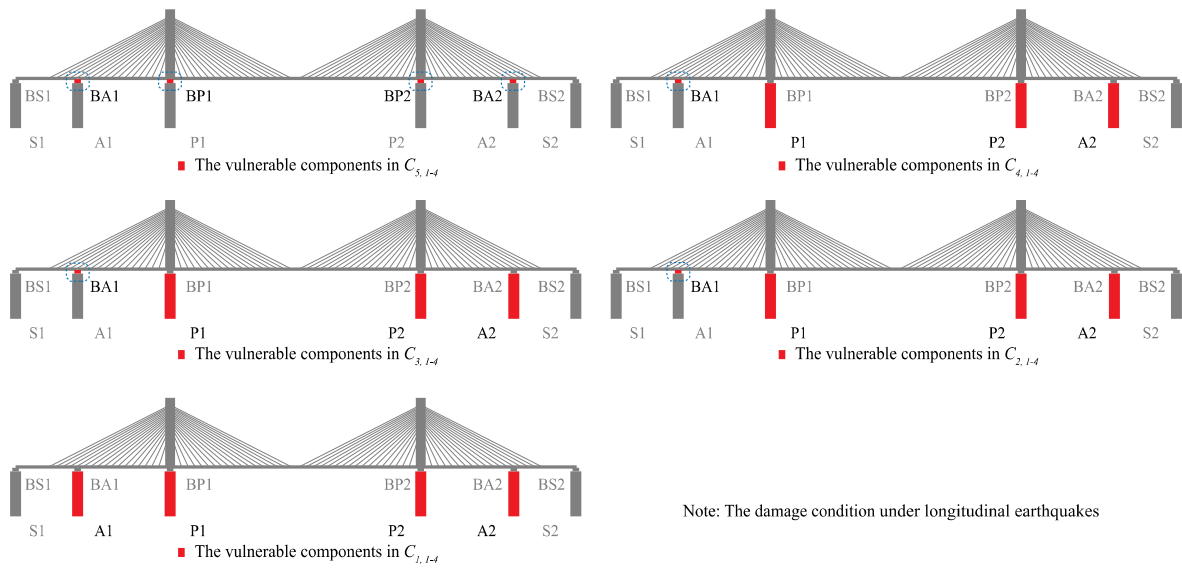


Fig. 16. Illustration of the change in location of vulnerable components for each case under MD.

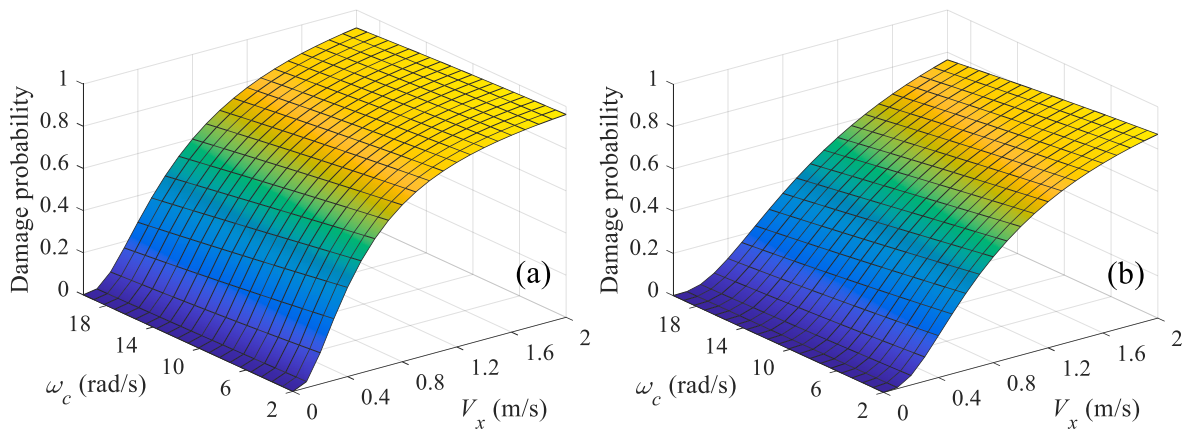


Fig. 17. Fragility surface of P1 for different damage states.

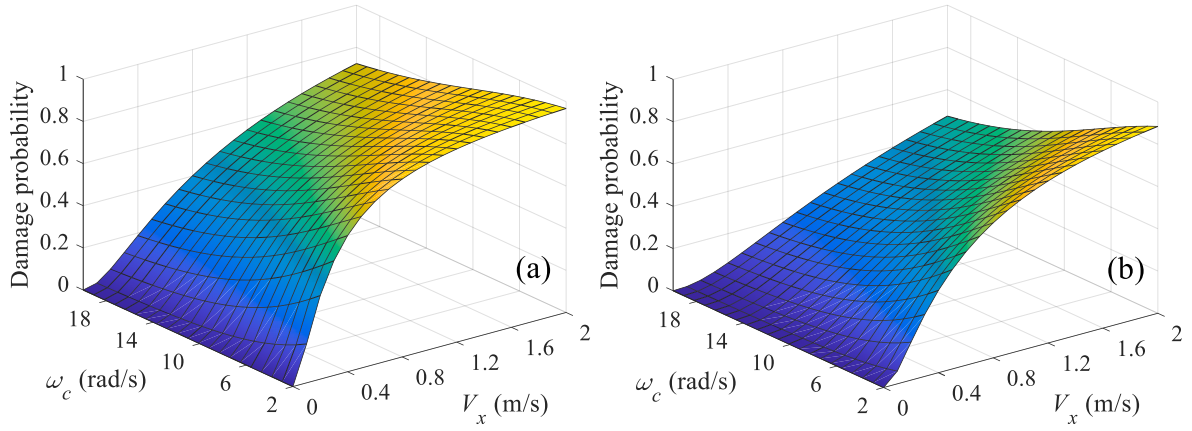


Fig. 18. Fragility surface of BP1 for different damage states.

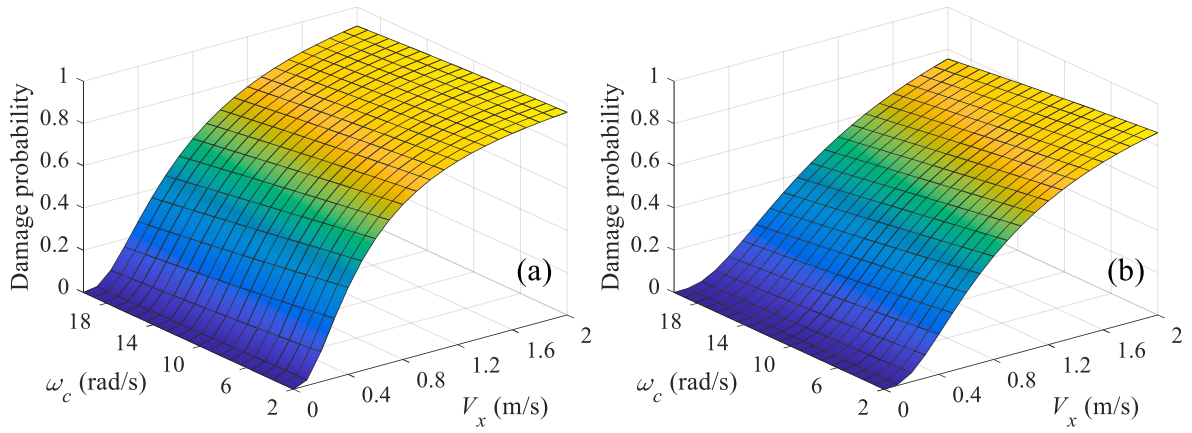


Fig. 19. Fragility surface of P2 for different damage states.

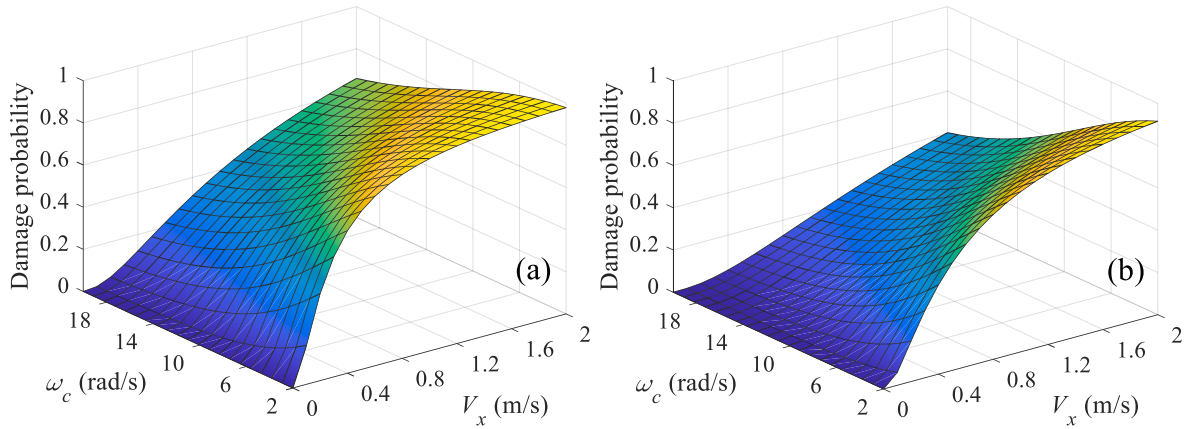


Fig. 20. Fragility surface of BP2 for different damage states.

combined action of V_x and ω_c , whereas region R_{P1} indicates that $P_{f-DP1} > 0$, i.e., pylon No. 1 is more vulnerable than its corresponding bearing with the increase of V_x and ω_c .

In the same way, the difference value of the damage probability between pylon No. 2 and its corresponding bearing under the joint participation of V_x and ω_c can be obtained with

$$P_{f-DP2}(V_x, \omega_c) = P_{f-P2}(V_x, \omega_c) - P_{f-BP2}(V_x, \omega_c) \quad (19)$$

Then, two regions (R_{P2} and R_{BP2}) divided by $P_{f-DP2} = 0$ associated with SD and MD are developed as depicted in Fig. 22(a) and (b),

respectively. In which region R_{BP2} ($P_{f-DP2} < 0$) implies that the bearing of pylon No. 2 is more vulnerable than pylon No. 2 under the combined action of V_x and ω_c , and region R_{P2} ($P_{f-DP2} > 0$) implies that pylon No. 2 is more vulnerable with the increase of V_x and ω_c .

Similarly, the damage probability difference (P_{f-DA1} and P_{f-DA2}) between auxiliary piers No. 1–2 and their corresponding bearings under the combination of V_x and ω_c is easily given with

$$P_{f-DA1(2)}(V_x, \omega_c) = P_{f-A1(2)}(V_x, \omega_c) - P_{f-BA1(2)}(V_x, \omega_c) \quad (20)$$

Immediately thereafter, different regions divided by $P_{f-DA1(2)} =$

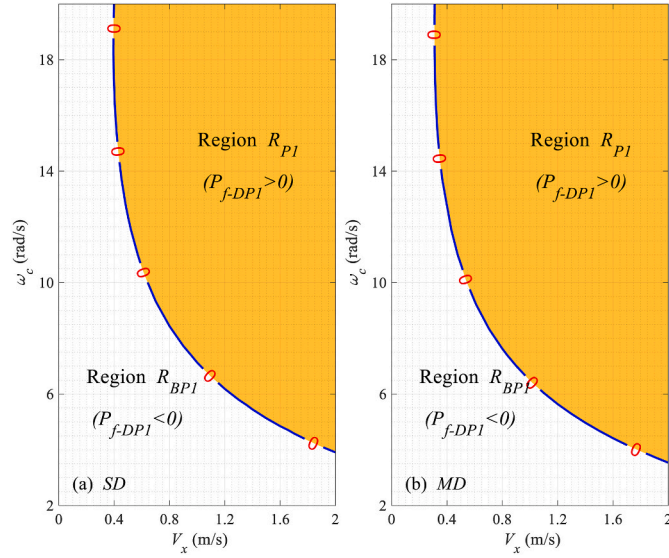


Fig. 21. Damage probability difference value (P_{f-D}) of P1 and BP1 for: (a) SD, (b) MD.

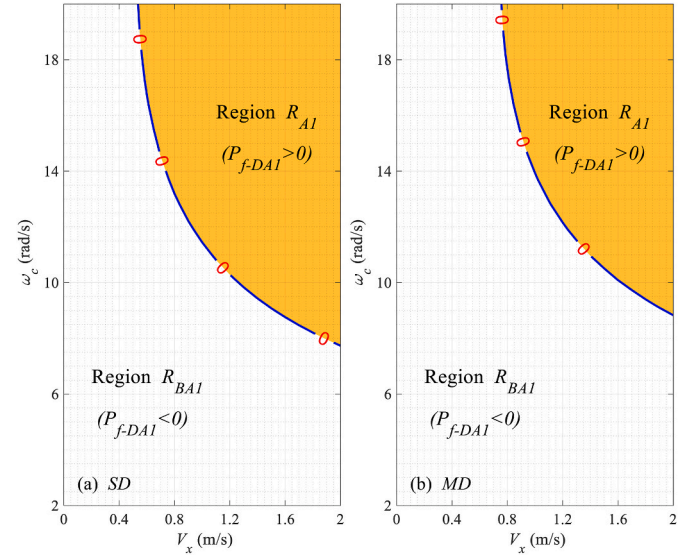


Fig. 23. Damage probability difference value (P_{f-D}) of A1 and BA1 for: (a) SD, (b) MD.

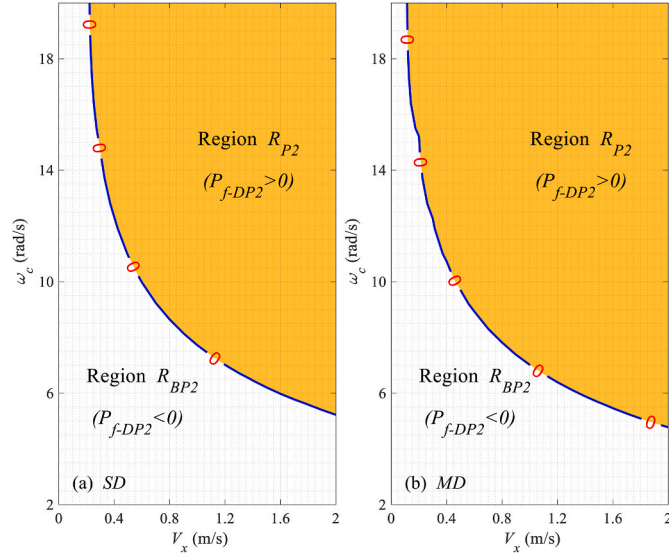


Fig. 22. Damage probability difference value (P_{f-D}) of P2 and BP2 for: (a) SD, (b) MD.

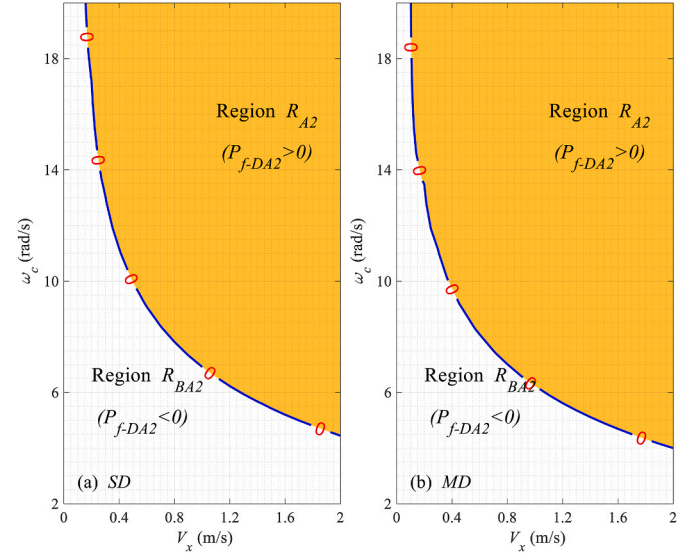


Fig. 24. Damage probability difference value (P_{f-D}) of A2 and BA2 for: (a) SD, (b) MD.

0 associated with SD and MD are developed as shown in Figs. 23 and 24, respectively. In which region R_{BA1} ($P_{f-DA1} < 0$) shown in Fig. 23 implies that the bearing of auxiliary pier No. 1 is more vulnerable than auxiliary pier No. 1 under the combination of V_x and ω_c , and region R_{A1} ($P_{f-DA1} > 0$) implies that auxiliary pier No. 1 is more vulnerable with the increase of V_x and ω_c . Region R_{BA2} ($P_{f-DA2} < 0$) shown in Fig. 24 implies that the bearing of auxiliary pier No. 2 is more vulnerable than auxiliary pier No. 2, while region R_{A2} ($P_{f-DA2} > 0$) implies that auxiliary pier No. 2 is more vulnerable under the joint participation of V_x and ω_c .

Following the above analysis and discussion, the damage intervals corresponding to different components in different intervals of parameters (V_x and ω_c) are finally gained, and each interval indicates that the corresponding component is the vulnerable component in that interval. Thus, the influence of the site-response effect on the failure mode and damage mechanism of long-span cable-stayed bridges are quantified and investigated with different combinations of seismic parameters V_x and ω_c .

6. Conclusions

Long-span cable-stayed bridge is a common bridge type for sea-crossing passages, and occupies an essential role in the transportation system. The spatial effects of the ground motion can remarkably impact the seismic response and damage sequence of bridge components. Therefore, a long-span sea-crossing cable-stayed bridge with consideration of hydrodynamic and pile-soil effects is employed to explore mainly the influence of the site-response effect on the damage sensitivity of critical components. The seismic fragility analysis of critical components of the cable-stayed bridge is carried out in Sections 5.1 and 5.2 to evaluate the damage condition of the piers, pylons and bearings under complex spatial variability earthquakes. Finally, the influence of the site parameter ω_c on the failure mode of cable-stayed bridges is investigated and quantified in Section 5.3. The following conclusions are gained.

- (1) From the observation of the fragility curves of pylons, piers and bearings under the action of complex spatial variability ground motions, it could be concluded that after the soil type at the pylon changes from soft to solid, auxiliary piers No. 1–2 and pylons No. 1–2 are more vulnerable than bearings at these corresponding locations under slight and moderate damage states, which exhibits different damage mechanism and failure mode, causing auxiliary piers and pylons to become vulnerable components.
- (2) On the basis of ω_c (the natural frequency of the site c) and V_x (the horizontal peak ground velocity), the seismic fragility surfaces of pylons, piers and bearings with consideration of non-uniform excitations are developed. The results indicate that when V_x reaches a specific value, the probability of damage to auxiliary piers No. 1–2 and pylons No. 1–2 gradually exceeds that of the bearings at these corresponding locations as ω_c increases from 2 rad/s to 20 rad/s under *SD* and *MD*.
- (3) The effect of ω_c (the soil type) on the damage sequence of components and failure mode of long-span sea-crossing cable-stayed bridges is quantified with the difference of the damage probability. It is concluded that regions R_{BP1} , R_{BP2} , R_{BA1} and R_{BA2} imply that bearings at pylons No. 1–2 and auxiliary piers No. 1–2 are vulnerable components in their corresponding regions, respectively. Whereas the damage mechanism will change with the soil type at the pylon changes from soft to solid, regions R_{P1} , R_{P2} , R_{A1} and R_{A2} imply that pylons No. 1–2 and auxiliary piers No. 1–2 are vulnerable components in their corresponding regions, respectively.

Bridge structures are usually subjected to three-dimensional ground motions, and the seawater layer significantly affects the vertical component of ground motions, while this study only focuses on the dynamic response of the long-span cable-stayed bridge under

longitudinal ground motions. In addition, the spatial effects of ground motions also include the wave-passage and incoherence effects, which may significantly affect the response of structures. Therefore, future research could include recommendations such as a systematic study of the wave-passage and incoherence effects on the component response and damage mechanism of the sea-crossing cable-stayed bridge under three-dimensional ground motions.

CRediT authorship contribution statement

Jian Zhong: Conceptualization, Methodology, Supervision, Writing – review & editing. **Wei Xu:** Methodology, Software, Data curation, Validation, Writing – original draft. **Kai Wei:** Resources, Validation, Investigation, Project administration, Visualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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APPENDIX A

Table A

Characteristics for selected ground motions from the PEER NGA

-	Earthquake Name	Year	PGV (m/s)	M	R (km)	Z _{TOR} (km)	T _p (s)	V _{s30} (m/s)	d (km)	θ (°)
EQ1	San Fernando	1971	1.14	6.61	1.81	0	1.64	2016	21.9	7.5
EQ2	Imperial Valley	1979	0.35	6.53	0.65	0	2.34	242	10	3.8
EQ3		1979	0.73	6.53	0.07	0	3.42	265	19.4	5.4
EQ4		1979	0.51	6.53	8.60	0	4.52	203	26.9	21.1
EQ5		1979	0.49	6.53	3.95	0	4.13	206	27.7	4.7
EQ6		1979	0.67	6.53	1.35	0	3.77	203	27.5	0.8
EQ7		1979	0.52	6.53	0.56	0	4.38	211	27.5	4.8
EQ8		1979	0.76	6.53	5.09	0	6.27	202	26.4	14.6
EQ9		1979	0.53	6.53	7.50	0	4.82	203	18.9	17.2
EQ10	Morgan Hill	1984	0.53	6.19	0.53	0.5	1.07	561	24.6	0.4
EQ11		1984	0.11	6.19	9.87	0.5	1.23	663	26.5	1
EQ12	San Salvador	1986	0.80	5.80	6.30	5.9	0.81	489	6	12.1
EQ13		1986	0.56	5.80	6.99	5.9	1.13	456	6	12.8
EQ14	Superstition Hills	1987	0.18	6.54	18.48	0	2.13	266	9	64
EQ15		1987	1.34	6.54	0.95	0	2.39	349	16	3.4
EQ16	Loma Prieta	1989	0.43	6.93	10.97	3.9	1.64	309	20	30.9
EQ17		1989	0.35	6.93	11.07	3.9	1.73	271	20	25.4
EQ18		1989	0.36	6.93	12.82	3.9	2.64	350	20	27.1
EQ19		1989	0.42	6.93	8.50	3.9	4.57	381	20	23.2
EQ20		1989	0.42	6.93	9.31	3.9	5.65	348	20	27.6
EQ21		1989	0.86	6.93	5.02	3.9	1.57	1070	18.6	23.9
EQ22	Landers	1992	1.33	7.28	2.19	0	5.12	1369	47.5	20.2
EQ23		1992	0.51	7.28	23.62	0	7.50	354	65.8	17.2
EQ24	Northridge	1994	1.11	6.69	5.43	5	3.16	373	19.4	13.7
EQ25		1994	0.76	6.69	5.43	5	3.54	526	19.4	13.7
EQ26		1994	0.78	6.69	8.44	5	0.93	380	17.3	26
EQ27		1994	0.75	6.69	5.92	5	1.62	629	19.4	16
EQ28		1994	0.76	6.69	5.92	5	1.37	269	19.4	4
EQ29		1994	1.18	6.69	5.48	5	2.98	286	19.4	11
EQ30		1994	0.45	6.69	7.01	5	0.59	2016	19.4	1.5

(continued on next page)

Table A (continued)

-	Earthquake Name	Year	PGV (m/s)	M	R (km)	Z _{TOR} (km)	T _p (s)	V _{s30} (m/s)	d (km)	θ (°)
EQ31	Kobe	1994	0.56	6.69	7.01	5	0.84	2016	19.4	1.5
EQ32		1994	0.31	6.69	7.26	5	0.73	508	19.4	5.4
EQ33		1994	0.76	6.69	7.46	5	1.23	326	19.4	0.8
EQ34		1994	1.48	6.69	6.50	5	1.25	282	19.4	18.3
EQ35		1994	1.16	6.69	5.35	5	2.98	251	19.4	13.3
EQ36		1994	1.21	6.69	5.19	5	3.53	371	19.4	12.2
EQ37		1994	0.78	6.69	5.30	5	2.44	441	19.4	6.3
EQ38		1995	0.91	6.90	0.96	0.2	1.09	312	18.1	7.8
EQ39		1995	0.91	6.90	3.31	0.2	2.83	198	18.6	14.6
EQ40		1995	0.68	6.90	0.27	0.2	1.81	312	38.6	2
EQ41	Chi-Chi	1995	1.21	6.90	1.47	0.2	1.55	256	17.4	13.5
EQ42		1999	0.51	7.62	9.62	0	6.65	428	17.2	67.4
EQ43		1999	0.65	7.62	9.94	0	5.34	259	33.6	58.7
EQ44		1999	0.37	7.62	28.04	0	5.29	407	56	30.9
EQ45		1999	0.58	7.62	19.83	0	5.38	478	45.3	39.1
EQ46		1999	0.57	7.62	25.42	0	9.58	298	45.3	39.8
EQ47		1999	0.55	7.62	19.89	0	9.33	541	56	29.8
EQ48		1999	0.57	7.62	22.06	0	6.43	362	45.3	43.3
EQ49		1999	0.50	7.62	26.00	0	9.34	705	56	30
EQ50		1999	0.29	7.62	16.74	0	8.04	466	56	24.4

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